# Ep. 378: Apurv Jain on Alternative Data, Real-Time Macro Nowcasting, and AI's Economic Role

**Podcast:** Macro hive

# AI Executive Summary

1. Core Thesis MacroX proposes that the global economy is best understood through a real-time measurement layer rather than lagging government forecasts. By aggregating over 1,000 diverse alternative data pipelinesâ€”including social sentiment, satellite imagery, and transactional recordsâ€”MacroX measures the underlying state of growth, inflation, and labor bargaining power without relying on noisy or revised official statistics. This approach claims to offer a two-to-three-month lead time in nowcasting economic conditions, allowing investors to detect shifts in market sentiment and structural economic changes (such as the divergence between CEO and worker sentiment during the 2023 recession scare) before they appear in traditional data releases like GDP or CPI.

2. Key Arguments & Data

3. **Measurement vs. Forecasting:** Official nowcasts (e.g., Atlanta Fed GDP) are retrospective and suffer from sampling issues, often showing contradictory trends within a quarter. MacroX prioritizes measuring the current state of economic phenomena directly rather than forecasting based on past models.
4. **Real-Time Wage Pressure Tracking:** Inflation is driven largely by services inflation, which correlates with wage pressure. While official data uses lagged Employment Cost Index (2 months) or noisy CPI estimates, MacroX aggregates real-time signals such as food stamp usage, job search sentiment, and manufacturing nightlights to infer actual worker bargaining power.
5. **Multimodal AI Implementation:** Large Language Models are not used for generative chat but for structured feature extraction (word embeddings). Specific models like BERT may be preferred over newer LLMs for specific text tasks due to stability. The model combines text, image (CNNs), and structured data to normalize biases inherent in single sources like Twitter.
6. **Structural Breaks & Economic Reality:** During the 2023 recession scare, yield curve inversions signaled high recession probability (per

PMI and corporate sentiment). MacroX identified a divergence: while CEOs felt negative pressure on earnings, worker bargaining power remained high, suggesting the economy would not collapse. This highlights how diverse data sources prevent over-reliance on structural shock narratives.

7. **Official Data Reliability:** There is significant concern regarding declining response rates and funding for government surveys (e.g., BLS), increasing statistical variance. Despite this, MacroX acknowledges the need for coordination but prioritizes their proprietary data as more direct evidence of market reality.

8. **Investment Utility:** Changes in MacroX's nowcasts correlate with future performance across equities, FX, commodities, and bonds. Investors can use these metrics to position portfolios 1-3 months ahead based on underlying economic conditions rather than macro narratives.

9. Actionable Takeaways

10. **For Portfolio Managers:** Integrate real-time alternative data pipelines into your risk management framework. Do not rely solely on lagging indicators (like NFP or PMI); instead, measure current economic state variables like wage pressure and consumer sentiment to gain a time lead on market positioning.
11. **For Investors:** Understand that market dislocations often reflect underlying realities revealed by broad datasets. If official data suggests volatility but alternative data shows stability (e.g., workers vs. CEOs), consider the broader economic context before reacting to panic signals from yield curves or PMI.
12. **For Career Seekers & Graduates:** Break out of professional echo chambers by traveling to experience different environments. Focus on delivering deep, measurable results on specific problems rather than generic networking; demonstrate initiative through tangible project execution, as this proves value in an AI-enhanced talent market.

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# Full Transcript

Welcome to the MacroHive Conversations podcast. I'm Bilal Hafiz, head of strategy at MacroHive. Before we dive in, a quick update. Following MacroHive's acquisition by the BGC Group, BGC has launched a new regulated business called MacroHive Markets. For the first time, onboarded clients can also access execution services via MacroHive Markets. This new workflow will allow macro institutional investors to benefit from strategy and quant insights, alongside access to global liquidity. Please contact Andrew Simon or Gail Mojibriyan

Bloomberg or click the link in the description below for more details on Macrite markets and its service offering. Now, on this episode's guest, Apov Jayne. Apov is the founder and CEO of MacroX, a San Francisco-based startup building a real-time measurement layer for the global economy using alternative data. Before MacroX, Apov built machine learning, investing systems at Microsoft, traded FX derivatives at Deutsche Bank and worked at Bridgewater Associates. His research experience spans Microsoft Research and Harvard Business School with publications and invited talks across academia, central banks, and global forums, including Davos and the World AI Summit. He holds a degree from Georgia Tech and the University of Chicago, and is a passionate about mixed martial arts. Now, on to our conversation. Greetings and welcome. Apple is fantastic to have you on the podcast show.

It's great to be here, Blau. I've been looking forward to this for a while. And I've been really impressed by the work that you've been doing at MacroX. So perhaps what we can do to start with is if you just explain what you guys are doing there, then from there, we can go into the details. Sounds good. So at MacroX, we use many different kinds of alternative data to measure the economic state. So in other words, understand What is growth doing right now? What is inflation doing right now? What is the consumer doing right now? What will be the central bank? What is the labor market doing right now? So we actually map the actual state of the economy, obviously, because government data is one to three months lagged. So by the time

experts figure out what is really going on, it can be a little bit late for acting on it.

typically happens is that you have economists often working at banks and other independent research houses. And what they do is they generally speaking track economic data, official economic data releases, say for the labor market, the Farm Payrolls Report for the US. And then other survey data like the PMIs or jobless claims, and then they come up with a view around that all. So how does your approach differ from that? Is it that you're using that same data, but you have some extra magic algorithms on top of that? Or is this something else you're doing? Great question. So I'll give you an example. We have been concerned about inflation, second order effects, and now we're actually seeing central banks act on it. So one of the things that drives inflation, obviously services inflation, a big piece as you and I know about two thirds of inflation, especially in developed markets. And what in turn drives services inflation is wage pressure?

And now the problem is, if you are an economist at a bank, like we were, you have to infer that. Like you can either look at the noisy services CPI, that's a realization of like, hey, well, which services CPI is high? Was it the fact that we heroically imputed rents in October when the government shut down to zero? And then the CPI popped up because the rents came back in. So it's basically a noisy white line. It's a noisy line or the noisy CPI data that you get, or you can wait two months for the gold standard employment cost index. So that's the index that the BLS very carefully calibrates and that's the gold standard for like, okay, if that's going up, wage pressures are going up. But the problem is that's two months delayed. So what we do is we'll look at actual worker bargaining power.

with lots and lots of alternative data. So to give you an example, if workers are searching for food stamps, they're probably not doing very well. Their bargaining power is lower. Or they're tweeting, I can't find a job. Or satellite data shows us something. Or corporate earnings shows us something. So there's many sources that go in. We have 1,000 plus kind of pipelines across many data sources. So when that happens, you're like, oh, workers are feeling this way. That's the complex map of their mental state. And then employers, are they complaining about not being able to find specialist

workers and having to pay them extra bonuses and letting them work from from anywhere? So that was in 2022. And when they were complaining that much, so their bargaining power was lower, worker bargaining power was much higher. And now the situation is reversed. So the employers have much more bargaining power. Maybe I'll replace you with AI, you know, and workers have lower bargaining power. So we actually literally measure those things. And using that, we're able to say, hey, this is the actual wage pressure that earlier you just had to noisily model using that noisy services CPI and the lagged employment cost index. So our thing is available in real time. And the good thing is it tracks all the theoretical stuff that we grew up with, like vacancies over unemployment.

things like that, or the Fed-based book. So it just basically takes the same macro concepts that we're comfortable with, but because it measures them directly and in real time, it's just better. be very reductive here and say something like you're monitoring tweets around workers complaining about it's hard to find a job or something. How do you separate the issue of more people using Twitter or X than before? Is it just that more people are using it and so that's a reflection of more use of the platform rather than actual change in sentiment? How do you account for that? That's a great question and a tough one because at some level I mean, you can apply many kind of statistical techniques to normalize, look at the intensity of emotion. You can do page rank within Twitter and say, hey, these are the very expert users because they are in the network of and followed by people about a specific topic. So of course, there are many techniques you can do. But if you really step back, it's actually a difficult thing to fully correct for any biases in Twitter. So what we actually do is So sure we do all those things that I explained like we try to clean it the best we can but really what our approach is to say well let's have a thousand sources let's get all the blogs let's get all the news articles let's get all the searches let's get you know let's get all the actual traffic to malls let's get all the credit card data like Then it really starts to add up like not only are they tweeting that they didn't get a job, but they're actually spending less and they're searching for buying durable goods less. And then they're actually satellite data showing us that there's lower emissions or there's less nightlights. So in other words, hey, factories are manufacturing less so their concerns are valid. So it's more like a, we try to build a legal case by having lots and lots of different kinds of

evidence rather than say, rather than insist on any one evidence and say that, oh, we can magically perfectly clean it. I think that's a dubious claim, to be honest. So it sounds like you're using lots of data. So it's a big data problem. And it sounds multimodal in some ways too, because it sounds like you're using images as well. So from putting kind of geeky hats on, what AI are you using here? Great question. So for different types of data, different kind of techniques are more relevant. So for instance, that what we've used, like CNNs will be used for image data. So because images, that's what makes sense for them if you're using satellite data. And CNNs just for the benefit of our audience in case they don't know some of these terms. They're on that work. So yeah, they're a particular technique, Jan Le Koon pioneered who used to be at Facebook. So it's a cool technique to analyze essentially images. So you can do that or And then for text, now, obviously, you can use any of the modern LLMs. And occasionally, though, some of the older LLMs, or the grandfather LLMs, I'll give you a quick example. BERT, in some cases, can be more useful than, for instance, just using ChatGPT or Claude for specific things. So it actually varies by data type, which is why it took us about three and a half, four years to build.

just the infrastructure to handle those thousand different types of data and signals. So it's one of those things that making one of those pipelines takes a smart team a year. So they can be like, OK, I can read some papers. I can take all the corporate filings. It may be hard, but I can get them. I can actually try to understand what the sentiment is of what they feel about the economy or whatever. So you can basically make one of those. But one of those, as we discussed, is not enough because it could be that more corporations are now just talking about AI, just to signal things. There can be many biases. So that one kind of information is typically not enough.

Which I feel is what has led to not as much use, a serious use of alternative data and investing even today. That's just my view. Because one source has not been enough to generate, you know, alpha, so it's just hard to fully scale that for people. But we discovered that when you, instead of one, when you go to a thousand, you really combine all of them, that's when it is much more powerful. And I notice in your work, you typically the output is some kind of now casting estimates rather than forecasting. I mean, just the way you sort of track things. Why is that? Why do it that way then saying we're

forecasting? Yeah, it's a question we thought about and we get asked. So first, we feel that by measuring things correctly right now, as I mentioned, it already gives you a two month lead. So the quick example of that is our now cast.

actually forecast city surprises two months ahead. So changes that were now caused because we are already measuring things in real time. So in a sense, we're already ahead because the government data will refer to the last month and two months before. So it's already ends up being a forecast because you're like, well, we are real time and we understand wage pressures are here. So not only are we real time, but really a little bit ahead. And the government data is two, three months behind. So that gives us this three month lead. And what's really cool is consistent across 15 countries. So like the city surprise thing, it just works across the board. And that also applies to the widely followed official now cast estimates like Atlanta Fed has a now cast estimate of GDP. So does the New York Fed. And I, you know, there's some other central banks that do similar ones as well.

Yeah, I would actually make a bold claim. We are the next generation of beyond Atlanta Fed. I respect their work, and I've done some academic work, and I talk about them because they're pioneers, but they're not built for trading. Atlanta Fed GDP now, there are quarters in which it starts at, let's say, 2%, and it ends the quarter minus 2%. So if you run a quantitative model, if you take the last value, then you'd be like, oh, the entire time that Lena Fed was saying it's minus 2%, why didn't you go short the market? So when it began the quarter, it was saying it's great. It's 2.5%. Everything's fine. So that point in time problem, and that's a feature. It's not built for that. It's they're aware of it. They never advertise and say, oh, we're going to do that. It's just they take all the government data.

embedded into, they just said, what does it mean for GDP right now? If PMI came out like this, if non-farm payrolls came out like that. So as the data points come in, they just update their estimate of the GDP. That's what they're built for. But what we do is we just take the whole thing all at once. So we don't have to spend three months repeatedly updating the NAACAs because more data came in. We update because the economic phenomena changed. Like when liberation day happened, The first week over

knockouts fell a lot led mostly by animal spirit like things and the harder data like, you know, driving to mall for traffic cars, all that kind of stuff actually slowed down the next month. But what was interesting was the first week when the announcements came out, it dipped a lot. And then the next week it kind of moderated, then it's still moderated the week after. So we could actually see it in real time being like, oh, they're talking.

that the trade war back a little bit. So it was not a sampling issue because we had the whole sample. But each week, the change was actually because the rhetoric of the trade war was much more contained. So the market was not as worried. Yeah, that makes sense. And You mentioned earlier about the official data releases. So presumably you have to validate or cross check your output to the eventual data that does come out, whether it's the BLS, BEA in the US, ONS in the UK, and so on. So how does that work? And how do you deal with revisions that happen? So GDP gets revised a number of times in the US. What are you validating against? Is it the first print or the final revised print and dealing with those?

of issues. Yeah, great and thorny question. It's good that you're such a macro expert so you can ask exactly the pain point. So we actually find that we are better at the final number mostly because of what I explained because we're getting all the data up front. But as things come out, and once again, we try to cast a wide net so we won't just hang our hats on the GDP. We'll say, OK, if growth is high, what is the PMI doing? What is city surprises doing? What was NFP? Although that's extremely noisy. So what was the Chicago Feds Activity Index? So we'll once again cast a wide net and look at that. But then we do some careful validation. And we actually show that even if you use the first print or the last print for GDP, of our now cast or changes to our now cast adds value beyond that. So because it uses information, so there's a few different advantages. So one is the speed that we talked about. But the second is, do you have extra information, even if imagining that you don't have the speed? So sure, you have the speed, but not imagine, do you actually know something extra? And what we find is we actually know Because the data sources are so different than what are used in the official statistics, we still tend to be mostly survey based. So the data sources are so different. So we actually find that even if you took away the speed advantage, there's extra information in those data that we can

show that statistically. So back to your original question, the problem of revisions, you can't pretend it doesn't exist. So what we try to do is do our best and say, look, whether you'd look at That stuff is noisily converging to what we said three months ago. And what we mean by that is like, look, here's the GDP, sure it got revised. Let's look at both prints. They're both correlated, more correlated to the final print, which is nice. Then let's look at the PMI, let's look at other things. So once again, we end up building a bit more of a legal case, if you will, in a very statistical and systematic fashion. Then insisting that, nope, that one thing is correct because.

I feel like when you've seen the markets long enough, you try not to hang your hat on any one thing. You know, one issue with some of the official data, so take inflation, CPI, is that a lot of the data or components within there are not actually true measured prices. They're imputed in some way, or derived in some way, or there's a model that estimates it. Same with GDP to some extent as well. So are you trying to match that, you know, so somehow work out what their underlying imputed price methodology is, or you're trying to just look at your own measures, which could then lead to an ongoing basis, you know, between your estimates versus the official numbers. So what we do is, we do look at the underlying methodology, for instance, we wrote a deep dive, why is NFP revised so much? So we'll look at the birth death model and all those kind of things to say, hey, this could be. But we generally don't try to predict any one number. We don't actually predict the individual NFP releases decently because of revisions unless there's a big move. Then we'll be like, oh, there's a very big move coming. Then we have extra information. Because if you already know the state, the way it works is earlier when we were sat on the desk or when you look at all these government data come in, the analyst's mind is kind of trying to generate the state. It's basically looking at all these different data points that are coming, and then it will say, oh, Well, this data is a little bit noisy. This data was negative, but these 10 data points are quite positive. So I think it actually is robust growth. So it's basically our mind is doing that. So that's what it's based on. So that's what we already do and we try not to focus on. Oh, let's look at the problem in NFP with the birth death model and try to, you know.

try to guess what the QCW revisions will be, and then try to predict the NFP, and then also think about all the VR seasonal factors. So yes, you can do that. But if you know the state, what's interesting is the ultimate test is the market. So if you know the state and we look at changes to our nowcast, and then we say, oh, when the nowcast went up, basically stocks go up the next month. So asset prices contain a lot of information. So the fact that changes to other measures relate to changes to the market, the 10-year bond or inflation or commodities or the S&P. So that's another test of like, okay, then you really have information because if your data is predicting the markets, so that's useful versus trying to predict the very minute oddness of the measurement that the BLS does. But yeah, we do look at it.

We do have to calibrate against that data, so we can't completely ignore it. So we'll comment on it. We'll look at it. But we won't necessarily try to nail down exactly what that prediction would be because our output is not really like predictions of lots of different government data. Our output is, hey, here's growth. Here's the state of growth. Sure, we can comment on this time. There's some. you know something going on in that export there's this gold thing that Atlanta Fed had to do the gold without gold kind of GDP now so we can comment on that but and say hey this could be the reason why the government data is here our data is here and don't worry about it.

this is the real state of the economy. So we use it to explain, I would say sorry I went on and on, but we basically the synthesis is we use it to explain the deviations because people still use those data, but then we don't necessarily try to really nail them down individually for each data release. Okay, yeah. And in terms of stability of the model, I mean, obviously we have structural breaks, COVID shocks, sometimes certain alternative measures work really well for a period of time, then they, you know, stop working or aren't so useful anymore. How do you deal with all of that? And by extension, then, how often do you kind of re estimate everything, rebuild the models and so on? So those are actually great scenarios for us. Very counterintuitive. But I love structural breaks because we've set ourselves a humbler problem. We just say, let's just measure what's happening right now. Because structural break, in a sense, is an exposed judgment that experts make. And they say, oh, that was a structural break. Mostly what you don't see is people saying, we are in a structural break.

You're right, you mostly, that's an ex post, like when you statistically do stuff, you're like, wait, that data point in COVID looks totally different. I'm going to call that a structural break. And that's legitimate. That's not, there's nothing wrong with it. But the only thing is just like recessions that are identified ex post. But if you're just measuring things in real time, it's an easier problem because you're not trying to forecast for which you then need a very complex like, oh, this model needs to be fitted differently than those data points because the distribution of these data and data during COVID is entirely different. So then I'm going to call this a structural break. So that ends up being great for us. The other quick example in 2023, and this relates to your forecasting question also. So one of the indicators that we grew up.

like in grad school as well as you know practically is the yield curve right when the yield curve inverts you know mostly things are not going to be good for the economy but in 2023 that broke down so the yield curve inverted PMI came was plummeting and very respected economists called for 100 chance of recession and then we were just looking at our nowcast and we could see that while the corporates or the CEO types were not feeling good as much, mostly because they had to pay workers more. So the workers were feeling great. They had great bargaining power. And the CEO is not as much. And it seems that when you do the PMI survey, the purchasing managers are closer to management than they really are to workers.

And people have to use this because PMI is fast in manufacturing while a small sector, it's fast. And they make the tradeoff within speed and kind of breadth. And it's like, OK, PMI is a leading indicator. It's narrow, but it's fast. So I'm going to look at this. Then yield curve signals the same thing. OK, 100% tons of recession. But if the economy changed, like workers had extraordinary bargaining power post COVID. So it is a structural shock in a sense of like mostly the labor market and the economy move. together, but this time it really blew apart. So workers had huge bargaining power, CEOs not as much, so they're feeling terrible as reflected in our nowcasts. And then workers had great bargaining power as reflected in their higher wages. So the economy didn't collapse because we were looking at a wider data source. So we could actually say actually, yeah, it's not going gangbusters,

but it will be fine because we're seeing the activity being fine. So we could actually say that's not going to be a recession. And we had just come out of stealth and started posting some stuff. And one of our supporters and investors was, well, thank God it was not a recession. It would have been a recession for you because you were wrong. But we were not, we were just measuring the thing and we could just say, look, broad data shows, that's not as bad as the narrow data is making it seem. So I feel like that's just a way of explaining that structural break is very good for measurement as the 2023 example kind of showed that the labor market to stop behaving the same as they used to for the last 20 years. No, that's great. Now I wanted to go on to how you apply this to financial markets. But before we do that, I, you know, I do want to get some of your background as well to add some context to this.

So in terms of your background, you know, obviously you started sort of MacroX as your startup, but what were you doing before that? Well, you know, obviously I cut my teeth of working with you. I did engineering and stuff before. Then I went on and studied a lot of finance. And my first finance class was Professor Gene Fama, the father of Efficient Markets. And he said people come to my class and then they go out and try to beat the market. I don't understand why it was very funny, but amazing class. So I learned kind of from the expert kind of quants and Fama and then Cochran who has this famous factor zoo speech, which now I'm presenting on, you know, how to courier factor zoo. So kind of had this very quantitative finance grounding. Then I came over to Deutsche and then kind of learned how the FX markets work. And it was really cool to get into options also because I felt back then options were not as well understood by academia or I mean in terms of how they really trade. I mean of course they had the black souls.

formula and all this that came out of academia, but then all the deviations and how to really practically trade is a little bit different than writing a paper. So that was amazing to understand FX markets at Deutsche. And then I had some ideas of becoming a professor. We all have that at some point. Yeah, yeah, yeah. So then thought about that a little bit. Then the markets had changed me too much to just stick to more writing papers. I wanted to do something additional. They went off to Bridgewater, which

was very interesting to kind of get. I loved how Bob Prince and Ray Dalio kind of explained the markets and how what really their frameworks were. So it was just great and great to be there. And then I went all the way over to the West Coast at Microsoft there, Microsoft.

was the only firm that let me actually trade. I actually managed a 3 billion portfolio, which is very big for me at the time. Back then, the Eurozone was blowing up and I went to bed every night. Regardless of all the quantum models I'd built like Morgan Stanley, please don't default. But then Microsoft also let me be a senior scientist in Microsoft Research at the same time. I basically got to hang out with the future of all the ML, all the AI tools, all the cloud. then came out and like really applied that to macro. So yeah, it's a bit of a journey. Yeah. And I think it's been the perfect background for what you're doing now. And so the question then is, you know, giving you kind of know how the financial markets work and the players in the market. I mean, how would, you know, you suggest people, portfolio managers, risk takers, you know, economists, outside economists, how would they use your output?

So we were in the design partner stage and now we're broader. But even then we found that people kind of very traditional fundamental managers used it to position themselves kind of one, two, three months ahead because that's the lead our data provides. So they were not, some of them are really not interested in what exactly NFP would be, but more of what would be the labor market. Is it still low hire, lower fire, and hence slowly improving unemployment? So they care about that and what was the inflation state and then they can position their portfolio relative to that. So that gives the one to three month kind of understanding them. The large trend is really useful for them. And then some on the other hand are extremely quantitative. They don't care about our narratives or opinions. They just want to pull the data. And for them, they find that, you know, changes to our data predict returns very broadly across equities, effects, commodities, bonds.

So they just like to use that directly. So both options are available. What is useful or I think partly is also unique to us is the fact that it's the same thing. Like you can actually understand the quantitative factors that you're using. So it's not just, hey, which is a perfectly reasonable approach. Like, hey, we

took these hundred time series, some like oil inflation, like all things that we think were relevant. to be relevant to predicting inflation. This model has the lowest mean squared errors, like a very quantity approach. And then here's our prediction of next month's specific inflation. And that's fine. But you can't fully explain that. Like you can do things like you can say, I'm going to use SHAP, like I'm going to export things. Exactly. I'm going to say, what are the most important variable importance? So you can kind of interrogate the model and come to a little bit more comfortable comfort around it. But you can never fully say, oh, inflation is going to be a little bit higher because inflation expectations have noisily started moving up again. And wage pressures are doing this. And aggregate demand is doing that. Basically, how we would write is how we measure. So you can take that measurement and just do a work on it quantitatively. But if you wanted to talk about it or use it to think about your.

kind of one to three months or even longer view is also helpful. And just on that, I mean, you mentioned earlier that you are using large language models within your process. And so what we know about LLMs is they can hallucinate. There's also a stochastic element to them as well. So you aren't quite sure, you know, how stable the output is with the same, you know, data input. So how do you deal with that? So we deal it by being very targeted with how we use them because all the points you raised are valid and what happens to LLMs and then kind of averaging over them also same idea. So, but we will not use, we will not point an LLM and say, you know, or have a cloud workflow and say, oh, here's my data, go make me an outcast. Like that actually fails, not in an obvious sense because it can look pretty good, but be filled with kind of the issues you're talking about.

So what we'll do is like, well, let's take this text, and then I want you to generate features in this way. And then it's just a very specific, very contained thing. So it will take the text that I'm like, OK, your word embeddings, like generate that or generate these specific things. And then we're going to use that along with other things in the model. So that makes it much more contained. And then you can run it a few times to make sure that the output is similar. And then ultimately, it's back to the same because you're combining with many other data. So you're not really dependent on like one small mistake or one small thing can upset. Yeah, I mean, the

average will kind of wash it wash it out. And so what the model is currently telling us say about the US economy.

So the models are actually saying the US economy is fine. So the growth is above trend. And we have been seeing that for the last few months. So yeah, it's just very clearly above trend. And then earlier in the year, like with the war, there was these concerns around, oh, would it be potentially stacked inflation? What would happen? So we really didn't see any of that. And we still don't see it. But the growth is fine. I'm actually just pulling up the now cast to just actually tell you. And then on the inflation side, In the nation side, yeah, we see that the second order effects that everyone's concerned about are mostly not that high. So in other words, the corporate pricing intentions of the concerns about that are not so high, like the wage pressures are not so high. So we saw that a few months ago, we actually shared that even more broadly. And the fact that labor market is not a source of inflation, almost every single central bank I think is now actually acknowledging that services don't seem to be driving a big piece of inflation. So they came to the same kind of conclusion a little bit later. So we continue to see that. We do see inflation expectations like ticking up slightly once again, because obviously the energy concerns and the length of war, we actually see consumer anxiety is relatively elevated, especially in the US.

So this is consumer financial anxiety, like people's concerns about cannot afford my life. So this is very specific. So we see the consumer, the K-shaped economy is true. Like consumer anxiety is pretty elevated even though the aggregate income and spending is fine. And obviously we saw the strong retail sales. So our data is consistent with both aggregates are fine and anxiety for the bottom third is extremely elevated. So we see that. which feeds into their concerns about energy prices and other things. And is also not a good sign for the midterms, for the incumbents. Yeah. So when someone's anxiety is elevated, people want regime change. They want some change. Yeah, and I just wanted to ask a couple of bigger picture questions. One is there's been this concern about the reliability of official statistical data, whether that's to do with the lower response rates and surveys, whether the data is capturing the right segments of the economy with new segments emerging and all those sorts of things. What's your take on that?

Because it sounds like you have a much more direct perhaps even broader measure of what's actually going on. And so, you know, you could have a view then on actually the official data is not really matching what you're seeing. So paradoxically, kind of that way around. Yeah. So I am very concerned about the dropping response rates to the government data. And I mean, even in the UK, I think there was these labor surveys that had a 1,515% response rate and the alarms were raised.

So I think that's a worldwide problem. And it's not just me. I think serious people. So I had a great conversation with the former BLS commissioner, Erica McIntarfer. She's an expert in the field. She's also concerned. So the response rates are a genuine concern. And I think the situation is even more serious than I understood, which I learned from my conversation with her, that the lack of funding is also another problem. So the lack of funding, funding is drying up. And people who actually get us to respond to surveys, and the government doesn't pay you to respond to a survey, and it takes our time to respond to the survey. So there's really, you're like, do I really want to fill out this huge survey and explain that I am looking for a job? These are all very questions that you may not want to even tell your mom. So to make you respond, to those things for free is actually a non-trivial human skill. I didn't know that, so that's what Erica told me. And then these people, if you don't fund them and they're losing those people who can get people to respond. So it's actually even worse that people don't want to respond and now you're losing the people who could make people respond. So the response rates, unfortunately, seem like they will keep plummeting, which increases the variance. So it used to be that the variance, the actual variance in the data, just the pure statistical measure of the quality has declined. The variance has gone up, so the quality has declined. So this is all very real. There's nothing to do with the competence of the statisticians or the staff. It's just, if you cut the funding and people don't respond, there's very little. And they're all very dedicated people. There's very little they can do.

So that picture was worse than I had wanted it to be because even paradoxically, so this is another thing that we actually want the government data to exist because of kind of two reasons. One is as you talked about earlier, eventually, even if you had the best information and in some things we do, we track things like we make up like a initial jobless claims for the

UK, which doesn't exist. We basically take all the best statistics across the world, and then we track it for all economies. So we standardize it. So it's like, oh, if I could compare the initial jobless claims in the UK and US, that would be great, except UK doesn't report it. So we generate that actually. So in that way, we know. But so in my point is, even if we had this perfect information, but.

you still want coordination among people. So people will be like, how do I trust your data? Because eventually the government data should converge to your data or should match it. That's one. And second is trustworthy economic data. I think got a great achievement of macroeconomics. It makes the debates at least somewhat grounded in the same measurement. And you can draw different conclusions from the same measurement. And you can say, oh, this is happening because this politician did this wrong. thing but at least you have the same thing you can agree on and same with financial markets obviously so you can agree on something so the coordination job they provide they're released for free unfortunately macrox can't give out all its data for free as we are for profit so like you know for coordination for depth of financial markets like if people can't trust the data they will trade less so actually Even though our lead in the U.S. is the information edge, because the U.S. still has pretty good data, is not as high as let's say we've been in an emerging country, but the liquidity is much higher. So it actually works out better because even a smaller edge but applied to a giant market, you know, is much more lucrative because it enables, and the market's giant in part because.

You know, there are many reasons for the market to be big, but in part because you have trustworthy data that everybody agrees on. Yeah, I totally agree with you there. The big picture question I wanted to ask just in terms of AI. I mean, there's so much hype around this at the moment. There's, you know, people talking about AI has a non-trivial chance of destroying humanity. You know, we've had frontier lab guys talking about we need a coordinated effort to slow everything down. You know, we're close to AGI. You know, what's your take on all of this? I mean, no, these are very real concerns that I have kids and I worry about the world they will inherit. So I do worry about that. I mean, in some cases, the concerns may be overblown and a part of the inevitable height that comes with a great technological

advancement. So it's very clear that it's a great technological advancement. I think we're all agreed on that. It's a very significant technological advancement. Could you contain some of the risks with somewhat simple guardrails and best practices probably. Of course, some risks emerge because it's such a new technology that you've never thought about, absolutely. New risks are coming all the time. So... And just in terms of your process, say large language models, which I guess is the AI that we're all talking about. I mean, what has it allowed you to do that classical machine learning didn't allow you to do? Because you talk about BERT, you know, we had BERT forever and then FINBERT and...

We've had all the CNNs and gradient boost models, all the usual stuff that people use in machine learning. That combined with big data helped a lot after ImageNet. But what did LLMs themselves bring to the table? In your case, that the others didn't? I think, as with others, it allows us to scale very fast. So if you can take pieces of the thing that you, as I mentioned, so we actually apply produce strict risk controls to, you know, to make sure we're not sharing our D by P with, you know, all the recent issues about the navier stokes, but it does allow us one engineer to do really the job of five. So it's so we've seen that practically firsthand and So it allows us to do that. So it allows us to take things, give it pieces that we think are safe and it allows us to scale them very, very fast. So it essentially builds lots of Lego blocks very quickly and we're able to assemble them and not just.

Actually, one thing I found certainly is that it's allowed, at least in my teams, like the AI research guys to do engineering, which they weren't so good at before. So suddenly they can actually spin up a server and all of that kind of back-end stuff, which they have to pass on to the prod guys or to the network guys, they can just do it, which has been surprisingly a big gain for us. I didn't quite think about that in that way. It's incredibly empowering because I agree with you because the coordination Among teams is always a problem. In other words, getting the researcher and the engineer to really talk to each other and agree and essentially do the same thing is always filled with team unfrictions. But if the researcher can do your point, just like do 80% of what a great engineer would do with like a few lines of code, like few commands in natural language, like that's just

insanely empowering. It's like, they've got a top engineer on tap who will actually listen to them.

That's true. Now, you did mention about, you know, you have children and such and I asked this to all my guests. I mean, what advice would you give to people who have recently graduated? You know, we're coming after the summer or will graduate saying 20, 27, you know, and they're kind of looking ahead and getting really worried about entry level jobs. So one is I would say just travel a little bit if you have not here graduated. I know it seems very counterintuitive and everyone's extremely anxious. But if any of these words make you grace us with attention to this podcast, then I would say just if you can't please travel, it will really kind of break you out of this, oh, that's not the only thing. And that's just very hard to break out of unless you travel.

That's a great point. I think you can end up in this very depressing echo chamber where you're just in this rut of applying and just hearing terrible stories like AI taking your job back. Yeah, you're anxious the entire time. And then you go back. People are still just eating their soup and being calm, right? Huh? Is that even possible? OK, let me get out of SF. So one is just travel. Just travel a little bit. Restore some faith. at least it did for me. But it's more specifically and seriously, I would say do projects. And I don't mean like everybody says that. But what I mean is like, take a specific deep problem and try to find an expert who will guide you a little bit. I mean, honestly, now you have so much expertise on tab, right? You can just ask, but just take one problem.

that you think would be relevant for an expert or you see them discuss and just try to attack it very deeply. The great thing is now in three months, you can make enormous progress. So if you really do that and you actually deliver something that works and then you can, they will probably have a conversation with you because like even in other form, like we still need humans and we actually watch out for whoever comes up and it's like, Hey, I did this. amazing thing, which just shows it's all the same stuff. It's just supercharged. They showed their initiative. They tried to attack a difficult problem. And then they can produce something pretty amazing, so which actually could be applied to your point earlier, but all could be applied right

away almost. So then you're really proving value and initiative and you're able to have more tools to be able to do that. You have more tools even try to launch your own startup because the risks are much lower. So I mean, compete with us. I mean, not my interest. That is true. I mean, I think we're finally at the point of having one man startups. I mean, that was always this mythical thing, a one man startup. But now actually, it genuinely could be possible. Yeah. I mean, someone had like a billion dollar revenue, one person startup, then they hired their brother. So technically, apparently didn't remain because there was like, I'm just lonely. I just need to talk.

Okay, no, that's great. The other question I wanted to ask, and this again, I asked all my guests, is what are some of the books that have influenced you or maybe books recently that you've read that you'd want to recommend to our audience? So I started, as I mentioned, from this very kind of a classical strict Chicago style of training. So so far most book, you know, theory of financial markets or like, you know, those kind of book. which we had to read. So I was in fond of them then. But it actually did influence my thinking a lot because you're like, oh, Marcus are efficient. So it's not necessarily that I completely believe that, but it's more how he thought about it and how systematically he investigated it. So that definitely influenced me. And then, you know, moving on to and then I actually So Talib's book actually on dynamic hedging is a great book. I know he's got the full by randomness. And I think people should definitely read that. But he actually wrote, he was an options trader. He actually wrote a technical book on options trading, which before he became famous for kind of the full by randomness, which is applies very generally. So that was very helpful. And Soros, George Soros' books on reflexivity and stuff. So I really liked them.

because I feel like they did describe some of the phenomena or talked about bubbles or how markets can behave very differently than Fama's worldview, right? So he's like, this is efficient. He's like, not at all. And so, but then they both had great points that just helped me who was a beginner to kind of see, oh, you could think about it this way, or you could think about it entirely differently also in a very coherent fashion. So Soros and the Minsky on the, so Minsky at the whole. kind of Minsky moment and all those things. So I feel like, so that really influenced me. Then, you know, of

course, in literature, I've just always loved fiction and, you know, some biographies now, but I've always just loved fiction and comics. So fiction, like I actually, Siddharth, there's, you know, the book Herman has like Siddharth. That really influenced me. There's this phrase where you go, I can think, I can fast, I can wait. And for whatever reason, that just is always kind of struck with me. So when you try to do something difficult as you, that's great. Yeah. Yeah. Yeah. Now, final, I mean, final question. What's the best way for people to reach out to you, learn more about what MacroX does and connect with you? You know, you can reach out to me on LinkedIn, obviously, you can reach out to communications at macroxstudio.com. So you can just say, oh, you want to try our stuff, go to our website. So Yeah, any of those. Yeah, that's great. Yeah, yeah. I will add that to the link to the show note as well. So with that, it's fantastic speaking to you doing fantastic work, you know, very, you know, kind of original as well. So I do urge everyone to go check out your site and your work and keep up the good work. Well, thank you very much. I appreciate it.

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